



Southeast University

Introduction to Software Engineering

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Outline

Part 0. Overview

Part 1. Software Processes

Part 2. Modeling

Part 3. Quality Assurance

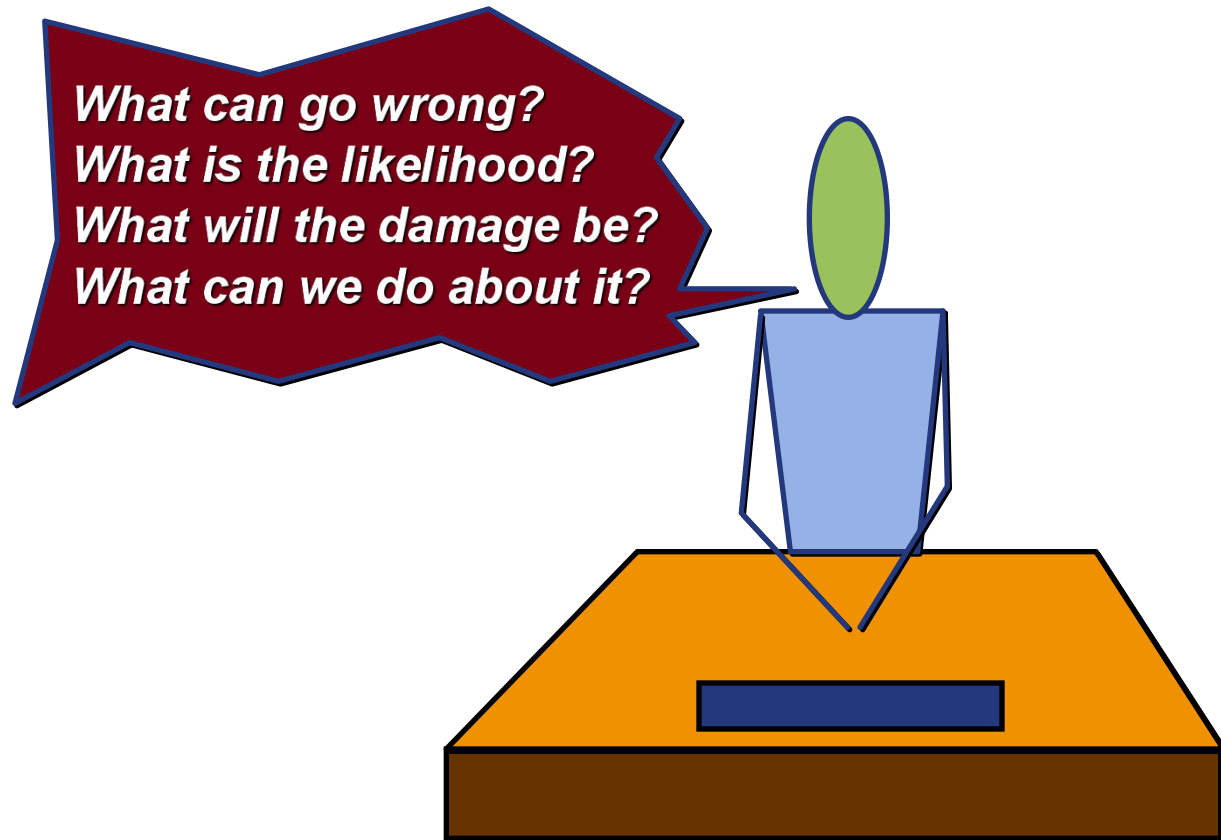
Part 4. Software Project Management

- ❖ **Chap 22. Project Management Concepts**
- ❖ **Chap 23. Process and Project Metrics**
- ❖ **Chap 24. Project Estimation & Scheduling**
- ❖ **Chap 26. Risk Analysis**

Chap 26. Risk Management

- ❖ **Project Risk**
- ❖ **Reactive Risk Management**
- ❖ **Proactive Risk Management**
 - **Risk identification**
 - **Risk analysis**
 - **Risk impact assessment**
 - **Risk projection**
 - **Risk planning**
 - **RMMM**
 - **Risk monitoring**

Project Risks



Risk Types

❖ Project Risk

❖ Technical Risk

❖ Business Risk

- Marketing risk
- Strategic risk
- Management risk
- Budget risk

- 风险类型
- 项目风险
- 技术风险
- 商业风险
 - 市场风险
 - 战略风险
 - 管理风险
 - 预算风险

Reactive Risk Management

- ❖ project team reacts to risks when they occur
 - **mitigation**—plan for additional resources in anticipation of fire fighting
 - **fix on failure**—resource are found and applied when the risk strikes
 - **crisis management**—failure does not respond to applied resources and project is in jeopardy （危险）
 - 项目团队在风险发生时做出反应
 - 缓解——计划额外资源以应对火灾
 - 故障修复——在风险发生时找到并应用资源
 - 危机管理——故障未响应应用资源，项目处于危急状态

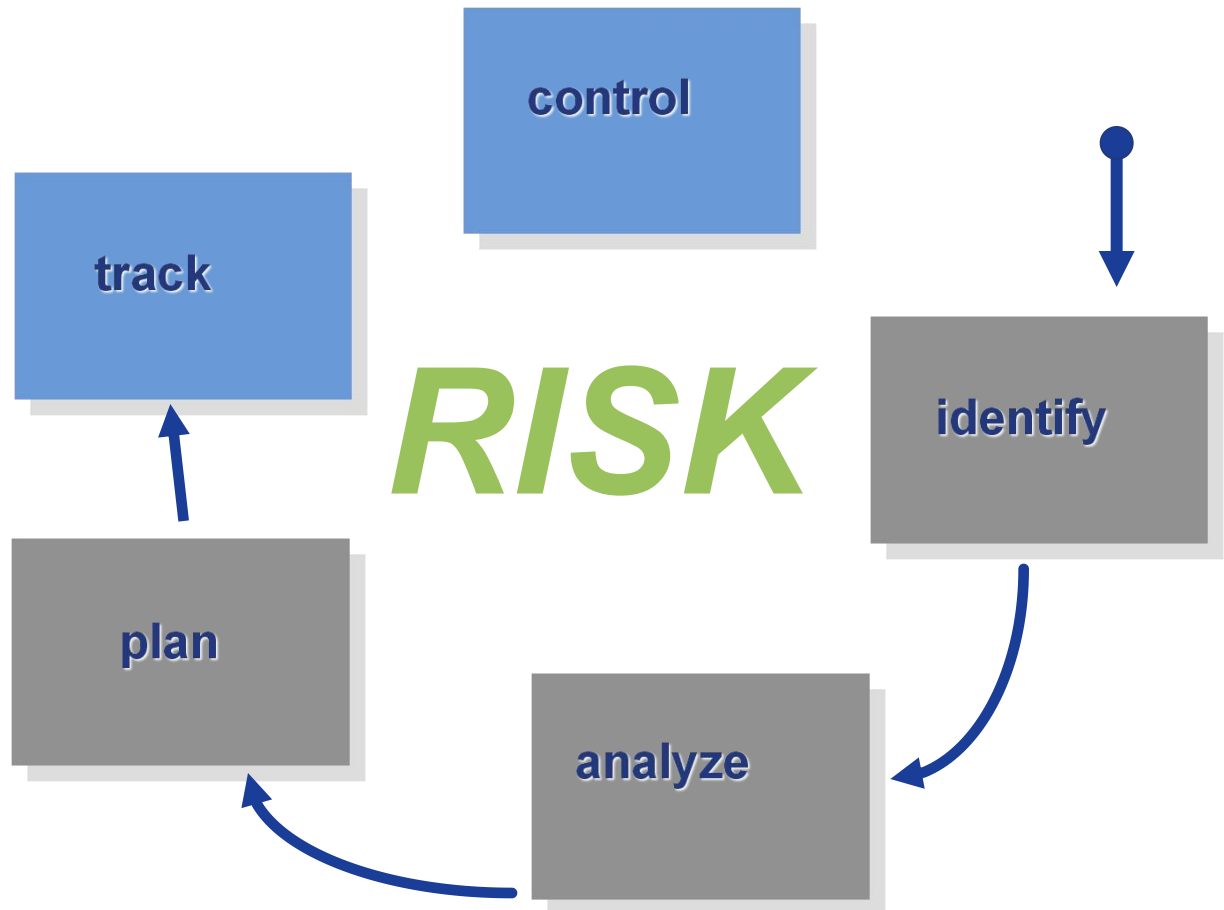
Proactive Risk Management

- ❖ **formal risk analysis is performed**
- ❖ **organization corrects the root causes of risk**
 - **TQM concepts and statistical SQA**
 - **examining risk sources that lie beyond the bounds of the software**
 - **developing the skill to manage change**
 - 进行正式的风险分析
 - 组织纠正风险的根本原因
 - TQM概念和统计SQA
 - 检查超出软件范围的风险来源
 - 发展管理变更的技能

Risk Management Paradigm 风险管理范式

- Risk identification
- Risk analysis
- Risk planning
- Risk monitoring

- 风险识别
- 风险分析
- 风险计划
- 风险监控



Risk Identification

- ❖ ***Product size***—risks associated with the overall size of the software to be built or modified.
- ❖ ***Business impact***—risks associated with constraints imposed by management or the marketplace.
- ❖ ***Customer characteristics***—risks associated with the sophistication of the customer and the developer's ability to communicate with the customer in a timely manner.
- ❖ ***Process definition***—risks associated with the degree to which the software process has been defined and is followed by the development organization.
- ❖ ***Development environment***—risks associated with the availability and quality of the tools to be used to build the product.
- ❖ ***Technology to be built***—risks associated with the complexity of the system to be built and the "newness" of the technology that is packaged by the system.
- ❖ ***Staff size and experience***—risks associated with the overall technical and project experience of the software engineers who will do the work.

Risk Identification

- 产品规模——与要构建或修改的软件整体规模相关的风险
- 商业影响——管理层或市场施加的约束相关的风险
- 客户特征——与客户的成熟度和开发人员按时与客户沟通的能力相关的风险
- 过程定义——与软件过程的定义和遵循程度相关的风险
- 开发环境——与构建产品所用工具的可用性和质量相关的风险
- 要构建的技术——与要构建的系统的复杂性和系统所采用技术的新颖性相关的风险
- 员工规模和经验——与执行工作的软件工程师的整体技术和项目经验相关的风险

Assessing Project Risk-I

评估项目风险

- ❖ **Have top software and customer managers formally committed to support the project?**
- ❖ **Are end-users enthusiastically committed to the project and the system/product to be built?**
- ❖ **Are requirements fully understood by the software engineering team and their customers?**
- ❖ **Have customers been involved fully in the definition of requirements?**
- ❖ **Do end-users have realistic expectations?**

Assessing Project Risk-II

- ❖ **Is project scope stable?**
- ❖ **Does the software engineering team have the right mix of skills?**
- ❖ **Are project requirements stable?**
- ❖ **Does the project team have experience with the technology to be implemented?**
- ❖ **Is the number of people on the project team adequate to do the job?**
- ❖ **Do all customer/user constituencies agree on the importance of the project and on the requirements for the system/product to be built?**

评估项目风险

- 软件和客户管理层是否正式承诺支持项目？
- 最终用户是否热情承诺项目和要构建的系统/产品？
- 软件工程团队及其客户是否完全理解需求？
- 客户是否完全参与需求的定义？
- 最终用户是否有现实的期望？
- 项目范围是否稳定？
- 软件工程团队是否具备合适的技能组合？
- 项目需求是否稳定？
- 项目团队是否有实施技术的经验？
- 项目团队的人数是否足够完成工作？
- 所有客户/用户群体是否同意项目的重要性和要构建的系统/产品的要求？

Risk Components

6. 风险因素

2. 成本风险

3. 支持风险

1. 性能风险

4. 进度风险

- ❖ ***performance risk***—the degree of uncertainty that the product will meet its requirements and be fit for its intended use.
- ❖ ***cost risk***—the degree of uncertainty that the project budget will be maintained.
- ❖ ***support risk***—the degree of uncertainty that the resultant software will be easy to correct, adapt, and enhance.
- ❖ ***schedule risk***—the degree of uncertainty that the project schedule will be maintained and that the product will be delivered on time.

Impact Assessment

components category		performance	support	cost	schedule
Catastrophic (灾难的)	1				
	2				
critical	1				
	2				
Marginal (轻微的)	1				
	2				
negligible	1				
	2				

Risk Projection 风险预测

- ❖ ***Risk projection*, also called *risk estimation*, attempts to rate each risk in two ways**
 - the likelihood or probability that the risk is real
 - the consequences of the problems associated with the risk, should it occur.
- ❖ **There are four risk projection steps:**
 - establish a scale that reflects the perceived likelihood of a risk
 - delineate the consequences of the risk
 - estimate the impact of the risk on the project and the product,
 - note the overall accuracy of the risk projection so that there will be no misunderstandings.

Risk Projection 风险预测

- 风险预测，也称为风险估算，尝试以两种方式评估每个风险
 - 风险是真实的可能性或概率
 - 风险发生时与之相关问题的后果
- 有四个风险预测步骤：
 - 建立反映风险感知可能性的量表
 - 描绘风险的后果
 - 估算风险对项目和产品的影响
 - 注意风险预测的总体准确性，以确保没有误解

Building a Risk Table

Risk	Probability	Impact	RMMM
			Risk Mitigation Monitoring & Management

Building the Risk Table

- ❖ Estimate the **probability** of occurrence
- ❖ Estimate the **impact** on the project on a scale of 1 to 5, where
 - 1 = low impact on project success
 - 5 = catastrophic impact on project success
- ❖ sort the table by probability and impact
 - 估计项目对项目的影响，范围为1到5，其中
 - 1 = 对项目成功的影响低
 - 5 = 对项目成功的灾难性影响
- 构建风险表
- 估计发生的概率
 - 按概率和影响对表进行排序

Risk Table

Risk	Category	Probability	Impact	RMMM
Larger number of users than planned	Product size	30%	3	...
...	...	...	...	...

Risk Exposure (Impact)

The overall *risk exposure*, RE , is determined using the following relationship [Hal98]:

$$RE = P \times C$$

where

P is the probability of occurrence for a risk, and

C is the cost to the project should the risk occur.

总体风险暴露 (RE) 通过以下关系确定 [Hal98]:

$$RE = P \times C$$

其中

P 是风险发生的概率,

C 是风险发生对项目的成本。

RMMM

- ❖ **Mitigation**—how can we avoid the risk?
- ❖ **Monitoring**—what factors can we track that will enable us to determine if the risk is becoming more or less likely?
- ❖ **Management**—what contingency plans do we have if the risk becomes a reality?
- ❖ **E.g. the RMMM for the risk of staff turnover**

- 缓解——如何避免风险?
- 监控——可以追踪哪些因素使我们能够确定风险是否变得更可能或更不可能?
- 管理——如果风险成为现实，我们有哪些应急计划?
- 例如，员工流失风险的RMMM

Recording Risk Information

Project: Embedded software for XYZ system

Risk type: schedule risk

Priority (1 low ... 5 critical): 4

Risk factor: Project completion will depend on tests which require hardware component under development. Hardware component delivery may be delayed

Probability: 60 %

Impact: Project completion will be delayed for each day that hardware is unavailable for use in software testing

Monitoring approach:

Scheduled milestone reviews with hardware group

Contingency plan:

Modification of testing strategy to accommodate delay using software simulation

Estimated resources: 6 additional person months beginning in July



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Thank You !

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